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BEng(Hons) Software Engineering

Module Name: 5COSC021C.2 Software Development Group Project

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Individual Report

"Attendify"

Team Number : SE 02

Project Title : Attendance Management System

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Declaration

I hereby declare that this project report and all the artifacts associated with it is my original work and has not been previously submitted or currently is being submitted for any academic program.

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Date: 07/01/2024

Abstract

This project's goal is to track and record students' and lecturers' attendance and engagement levels in the classroom while studying and identify beneficial ways and resources within the suggested system. The primary goal is to address the challenges that lecturers experience in enhancing student attendance and the whole process. This individual report focuses on improving the proposed attendance management system more effectively. The study uses a variety of requirement-eliciting techniques, such as interviews, questionnaires, brainstorming, etc., to seek a proper understanding of the requirements that are needed. The project uses RFID technology to track attendance. This study also contributes valuable insights on seamlessly tracking student engagement. The report will focus on system specifications, exploring social, legal, ethical, and professional issues that may arise when developing a project, and finally providing a detailed insight into the system architecture and design.

Keywords - Attendance, RFID, System Architecture

Acknowledgement

First and foremost, I would like to extend my deepest gratitude to my mentor and the module leader, Mr. Banuka Athuraliya, for his constant support, guidance, and support throughout the semester. I am grateful to him for advising me and my team and helping us move in the right direction when we encountered difficulties along the way. My sincere thanks also go to my colleagues who helped us by participating in the survey. Last but not least, I would like to acknowledge my team members for putting in their maximum effort, and I would also like to express my deepest gratitude to my friends and family members who assisted me in many ways.

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Abbreviations table

Abbreviation	Explanation
NFC	Near-field communication
RFID	Radio-frequency identification
SRS	System Requirements Specification
CSV	Comma separated values
BCS	British Computer society
PWA	Progressive Web Apps

Chapter 4: System Requirements Specification (SRS)

4.1 Chapter Overview

The System Requirements Specification (SRS) will focus on conducting a comprehensive stakeholder analysis, which will provide an in-depth idea on the people who will influence Attendify. The information gathered from the questionnaires and interviews are analysed followed by a discussion of selecting proper requirement elicitation techniques. Additionally, this chapter also identifies the use cases of the system. In conclusion, a detailed explanation of Attendify's functional and nonfunctional requirements are provided.

4.2 Stakeholder Analysis

4.2.1 Onion Model

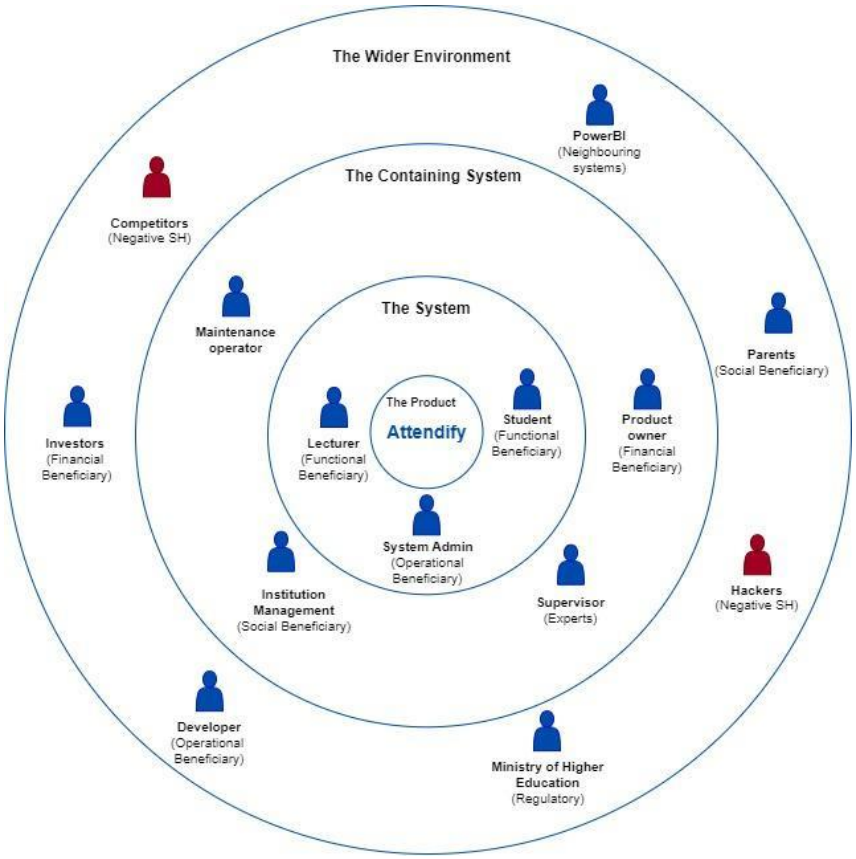


Figure 1 : The Onion Model (Self Composed)

4.2.2 Stakeholder Descriptions

Stakeholder	Viewpoint
Functional beneficiary	
Students	Will utilise the system to check their timetables and track their own attendance history.
Lecturers	Will utilise the system to check, update the timetables and view the attendance of the students and generate reports.
Financial beneficiary	
Product owner	Will earn profit by giving the system to educational institutions.
Investors	Will earn profit from the money that they get after selling or giving the systems to organisations.
Social beneficiary	
Parents	Wants to ensure that their child attends all the sessions, and the system will notify the parents if students do not attend lectures.
Institution Management	Wants to track and monitor the students attendance and involvement in lectures for effective management and academic performance.
Operational beneficiary	
System Admin	Wants to ensure that proper access levels are given to users and attendance is being tracked properly.

Developer	Wants to introduce new features and enhance the system in order to ensure a seamless user experience.
Negative Stakeholders	
Hackers	Wants to interrupt the flow of the system.
Competitors	Wants to build a better project than the proposed project.
Regulatory	
Ministry of Higher Education	Wants to ensure that students are involved in academic related activities in order to foster and uphold higher standards.
Experts	
Supervisor	Wants to provide comprehensive guidelines to implement the project successfully.
Neighbouring systems	
PowerBI	PowerBI will help the system to visualise and produce attendance related reports.

Table 1 : The Stakeholder descriptions table

4.3 Selection of Requirement Elicitation Techniques/Methods

Requirements gathering or requirements elicitation is a crucial phase that could determine the success of the project. During the research period, the team decided to use certain requirement gathering methods in order to collect data from identified stakeholders. The following techniques were chosen.

- Brainstorming
- Questionnaire

- Interviews
- Prototyping

The section below would describe more about the selected requirements gathering methods.

4.3.1 Brainstorming

Brainstorming is usually done during the initial phase of the research process, and it is often considered the starting point of the project. It fosters creative thinking and enables the opportunity to identify new ideas within the team. It is also significant to note that brainstorming Interviews ensures everyone in the team has a good understanding of the problem and the steps that are needed to solve it. Even though brainstorming has certain disadvantages, such as a lack of structure, being time-consuming, etc., since it involves every group member, many new ideas about system requirements can be gathered within a shorter period of time. Hence, it was necessary to use brainstorming as a requirement gathering technique.

4.3.2 Questionnaire

A questionnaire serves as a well-structured and systematic method to gather information from a larger audience within a short period of time. The questionnaire was made and sent to students in order to gather their opinion on the current attendance marking system and to get their opinion on the proposed solution. The team decided to enable the anonymous response option for the questionnaire in order to gather more information because respondents feel more comfortable answering when their responses are saved in an anonymous form. The compulsory questions were mostly about choosing an option. This encouraged the students to fill out the questionnaire; hence, it would only take a maximum of two minutes. The built-in analytical feature of Google Forms helped to easily categorize the students' opinions.

4.3.3 Interviews

Interviews often serve as a crucial part of the requirement elicitation process. Especially interviews with domain experts will provide more insight on details such as specific industry expertise, unique requirements etc. The knowledge that they possess in the specific field will be a crucial asset to build a seamless project. Despite the time it may take to schedule and conduct an interview, the outcome will be a huge aid for the team. The team decided to have an interview with a lecturer to

understand the problems a lecturer is facing due to the current attendance tracking method. The interview was held with Mr. Kushan Bhareti, a visiting lecturer at Informatics Institute of Technology.

4.3.4 Prototyping

Prototyping is usually done before development in order to get an overview of the system and how it will work. It will also help to provide a seamless user experience for the users. Creating the mockups in the early stages and getting feedback from the stakeholders will aid in developing the system. Prototyping also gives a thorough understanding to the team members, which will ensure that no errors are made during the implementation period.

4.4 Discussion/ Analysis of Results

4.4.1 Questionnaire analysis

The questionnaire analysis table has been moved to the Appendix, Section A.2, along with the screenshots of the survey form that was sent to gather information from students for more clarity.

4.4.2 Interview analysis

The interview analysis table has been moved to the Appendix, Section A.3, for more clarity.

4.4.3 Summary of findings

Findings	Questionnaire	Brainstorming	Interviews	Prototyping
Should be able to access previous attendance records	✅	✅	✅	✅
The system should be able handle more than 10000 users at a time	❌	✅	✅	❌
Should have an integrated timetable module	✅	❌	✅	✅

The lecturers should have the ability to generate analytical reports	✗	✓	✓	✗
The system should send low attendance warning emails.	✓	✓	✗	✗

Table 2 : The Summary of findings table

4.5 Use Case Diagram

The following use cases diagram will provide the basic understanding of how users will interact with the Attendify system.

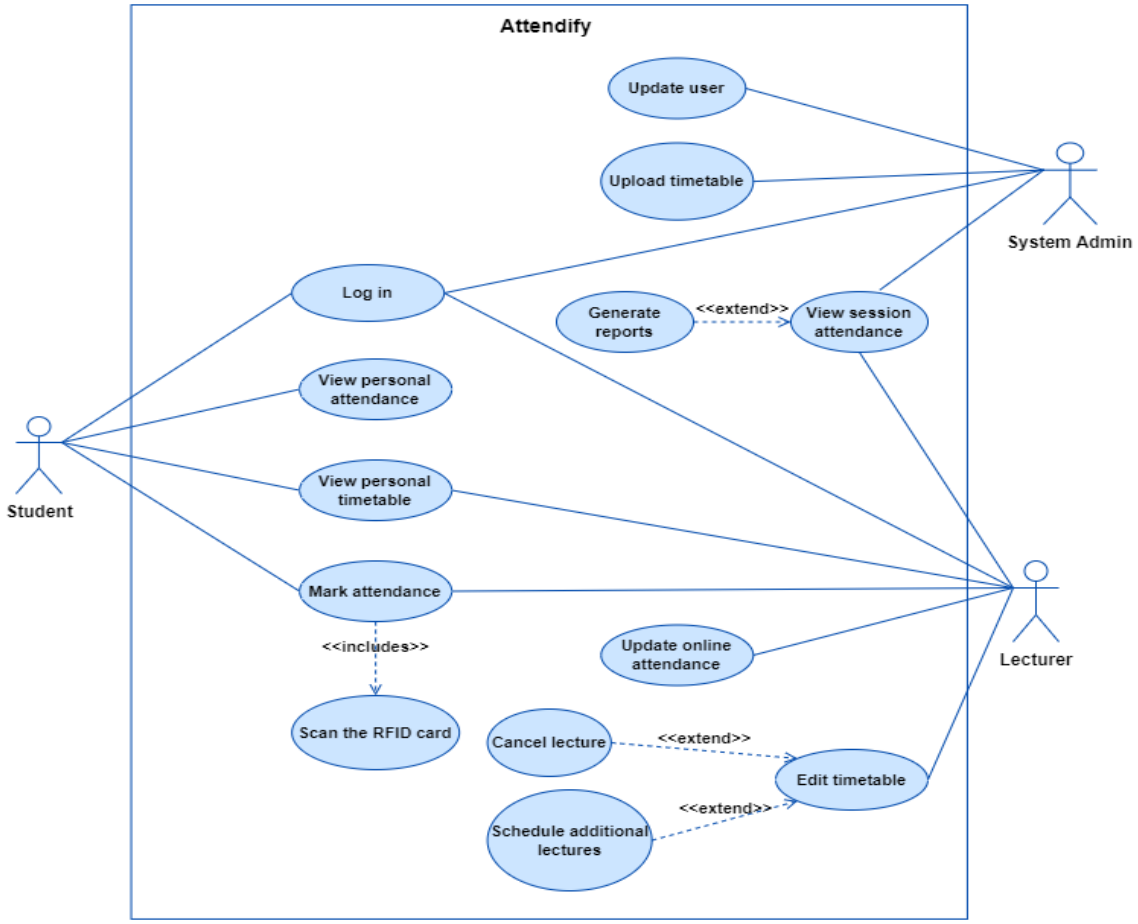


Figure 2 : Use case diagram(Self Composed)

4.6 Use Case Descriptions

The identified core use case descriptions are given below. Other use case descriptions can be found under Appendix Section A.4.

4.6.1 Mark attendance

Use Case Name	Mark attendance	
Use Case ID	UC-004	
Description	Users marking their personal attendance	
Priority	High	
Primary Actor	Lecturer, Student	
Supporting Actors	RFID Scanner	
Pre-Conditions	User should have a valid RFID card given from the organisation.	
Trigger	The user wants to mark their attendance.	
Main flow	**Actors**	**System**
	1) User scans the RFID card	2) System authorizes the scanned RFID card 3) System stores the attendance information
Exception flow	**Actors**	**System**
	None	None

Alternate flow	**Actors**	**System**
	1) User scans an invalid RFID card	2) System alerts invalid RFID card 3) System ends use case
Exclusions	None	
Post Conditions	The system stores the desired user's attendance.	

Table 3 : Use Case description - Mark Attendance

4.6.2 Update online attendance

Use Case Name	Update online attendance	
Use Case ID	UC-007	
Description	Lecturer uploading online attendance as CSV file to update attendance and also to store the attendance history.	
Priority	High	
Primary Actor	Lecturer	
Supporting Actors	Student	
Pre-Conditions	Lecturer should have a CSV file generated from the online session delivery platform.	
Trigger	The lecturer wants to update the online attendance record to the system.	
Main flow	**Actors**	**System**

	1) Lecturer selects update online attendance option 4) Lecturer enters the desired details 6) Lecturer uploads the csv file	2) System displays the update online attendance option page 3) System prompts the user to enter details regarding the session 5) System prompts the user to upload the CSV file 6) System updates the online attendance record
Exception flow	**Actors**	**System**
	1) Lecturer selects update online attendance option 3) Lecturer connects to the internet	2) System does not work due to bad internet connection 4) System continues to work in the desired manner
Alternate flow	**Actors**	**System**
	None	None
Exclusions	None	
Post Conditions	The system updates the online attendance according to the desired session.	

Table 4 : Use Case description - Update Online Attendance

4.6.3 View session attendance

Use Case Name	View session attendance
Use Case ID	UC-009

Description	Users who have access can view attendance for a particular session.	
Priority	High	
Primary Actor	Lecturer, Admin	
Supporting Actors	None	
Pre-Conditions	User should be logged into the system.	
Trigger	The user wants to view the attendance for a particular session.	
Main flow	**Actors**	**System**
	1) User selects view session attendance option 4) User enters desired information	2) System displays the view session attendance option page 3) System prompts the user to enter desired details regarding the session 5) System displays the desired attendance information
Exception flow	**Actors**	**System**
	1) User selects view session attendance option 3) User connects to the internet	2) System does not work due to bad internet connection 4) System continues to work in the desired manner
Alternate flow	**Actors**	**System**
	None	None

Exclusions	Generate reports
Post Conditions	The system displays the attendance for the selected particular session.

Table 5 : Use Case description - View Session Attendance

4.6.4 Generate reports

Use Case Name	Generate reports	
Use Case ID	UC-010	
Description	Users who have access can generate reports for particular sessions.	
Priority	High	
Primary Actor	Lecturer, Admin	
Supporting Actors	Student	
Pre-Conditions	User should be logged into the system.	
Trigger	The user wants to generate attendance reports for particular sessions.	
Main flow	**Actors**	**System**
	1) User selects generate session attendance option 3) User enters session details 5) User downloads the report	2) System prompts the user to enter desired details regarding the session 4) System displays download report option
Exception flow	**Actors**	**System**

	1) User selects generate session attendance report option 3) User connects to the internet	2) System does not work due to bad internet connection 4) System continues to work in the desired manner
Alternate flow	**Actors**	**System**
	None	None
Exclusions	None	
Post Conditions	The system generates an attendance report for a selected session.	

Table 6 : Use Case description - Generate Reports

4.7 Functional Requirements (with prioritization)

The following table will discuss the functional requirements of this project, as well as the priority level of each requirement.

- Critical – The requirements that are critically needed in the successful completion
- Desirable – The requirements that can add value, but are not required immediately
- Luxury – The requirements that would add luxury to the system

Requirements list		**Priority Level**	**Description**
FR1	Update user	Critical	The system should be able to update a particular user accordingly, as specified by the system admin.

FR2	Mark attendance	Critical	The system should be able to verify and store attendance seamlessly when the FRID scanner is triggered through the scanning of RFID cards.
FR3	Edit timetable	Desirable	The system should be able to reflect any changes made by a lecturer to the timetable by updating and notifying the students.
FR4	Update and store online attendance	Critical	The system should be able to convert and store the online attendance information that was uploaded as a CSV file by the lecturer.
FR5	View session attendance	Critical	The system should be able to display the requested session's attendance and should allow the user to generate reports if requested.
FR6	View personal attendance	Critical	The system should be able to display the personal attendance history of a specific user.
FR7	Upload timetable	Desirable	The system should be able to display the timetable that is being uploaded by the admin.

FR8	View personal timetable	Desirable	The system should be able to display the desired personal timetable when a user requests it.

Table 7 : Functional Requirements

4.8 Non-Functional Requirements

The following table will discuss the non-functional requirements of this project, as well as the priority level of each requirement.

- Critical – The requirements that are critically needed in the successful completion
- Desirable – The requirements that can add value, but are not required immediately
- Luxury – The requirements that would add luxury to the system

Requirements list		**Priority Level**	**Description**
NFR1	Performance	Critical	The system should function properly without any performance issues such as lagging, network latency, peak traffic handling etc.
NFR2	Accuracy	Critical	The system should be able to accurately store and retrieve information regarding attendance.
NFR3	Reliability	Critical	The system should be reliable for the institution management to store information related to students and lecturers attendance.

NFR4	Scalability	Desirable	The system should be scalable to effectively handle an increasing number of users in the future.
NFR5	Usability	Desirable	The user interface should be user friendly and smooth in order to provide a seamless user experience.

Table 8 : Non-Functional Requirements

4.9 Chapter Summary

This chapter focuses on identifying the suitable stakeholders for the proposed system, which was represented using an onion diagram followed by suitable descriptions. It also identifies the requirement elicitation methods, use case diagrams, use case descriptions and the functional and nonfunctional requirements with their respective priority levels.

Chapter 5: Social, Legal, Ethical and Professional Issues

5.1 Chapter Overview

This chapter will provide a comprehensive explanation of the responsibilities and challenges associated with building a project and how Attendify proactively aligns with the guidelines of the BCS code of conduct. Attendify has always ensured that it abides by ethical guidelines and embodies values and principles. This chapter will explain the importance of adhering to guidelines and provide an insight on how Attendify has balanced both technical innovation and the adherence to guidelines.

5.2 SLEP Issues and Mitigation

It is necessary to be aware of social, legal, ethical, and professional concerns while building software projects. The following topics will provide an understanding of how social, legal, ethical, and professional issues were managed according to the BCS code of conduct while building Attendify.

5.2.1 Social Issues

Project Attendify is an advanced attendance management system developed to eliminate problems related to attendance. This project is being developed for the purpose of helping the community, and it does not contain any harmful effects on society. Moreover, this project does not involve or represent any religion, politics, or ethnicity.

The initiative will only be implemented in the English language. This might cause challenges for non-native English speakers. However, the target market to implement the system is English-speaking educational institutions where English is their main and common language.

5.2.2 Legal Issues

This project does not involve any machine-learning components. Therefore, no datasets were used. The questionnaire sent during the requirement elicitation stage did not collect any personal or

sensitive information. The user's privacy was the first priority; therefore, data was collected anonymously and kept confidential.

The software used during the development period is either open source or legally licensed by the university. Finally, the team ensured that every piece of information acquired was given credit to the original author by referencing it using the Westminster Harvard referencing system.

5.2.3 Ethical Issues

The questionnaire used to gather information did not collect any email addresses or personal information. The domain expert who provided valuable insights through the interview was also informed how the information he provided would be used. All the information collected was solely used for the betterment of the project Attendify and will not be shared with any third party. Additionally, none of the software used or utilised in this project is cracked or pirated.

5.2.4 Professional Issues

The Attendify crew made sure that the project was developed by adhering to the BCS code of conduct and also by maintaining good professionalism. It is necessary to follow professionalism in order to make a project successful. All the information gathered was maintained confidentially by the Attendify crew. The team also made sure that all mentioned development, design and project management methodologies were followed.

5.3 Chapter Summary

This chapter discussed the social, legal, ethical, and professional issues that could arise during the development life cycle or even after customers start to use it. The chapter also outlined how the BCS code of conduct was followed in order to prevent any undetermined events that could occur. Furthermore, it emphasised the significance of taking proactive measures to ensure the success of the project.

Chapter 6: System Architecture & Design

6.1 Chapter Overview

This chapter will offer a comprehensive overview of the system architecture and design for Attendify. It will discuss the high level and low-level designs of the system. The chapter contains the system architecture design, class diagram, sequence diagram, activity diagram and finally the UI design, which will provide a visual representation of the frontend of the system.

6.2 System Architecture Design

The high-level architecture of Attendify is given below. Attendify's high level architecture is represented using three layers. The first one being the presentation layer which illustrates the web application and the RFID scanner. The second and middle is called the domain logic layer which represents the core functionalities of the system. Finally, the data storage layer which represents the data that has been stored in the database.

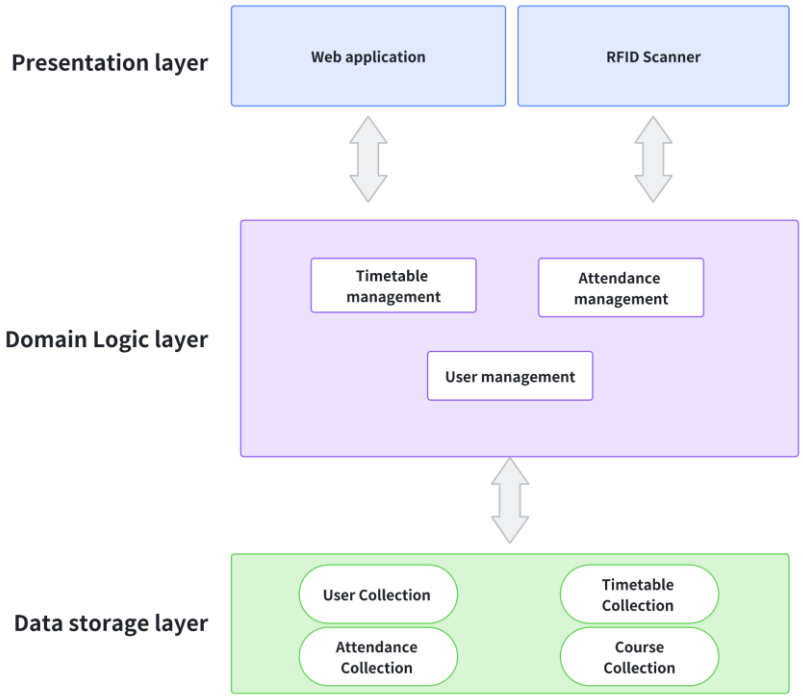


Figure 3 : High Level Architecture Diagram of Attendify

6.3 System Design

6.3.1 Class Diagram

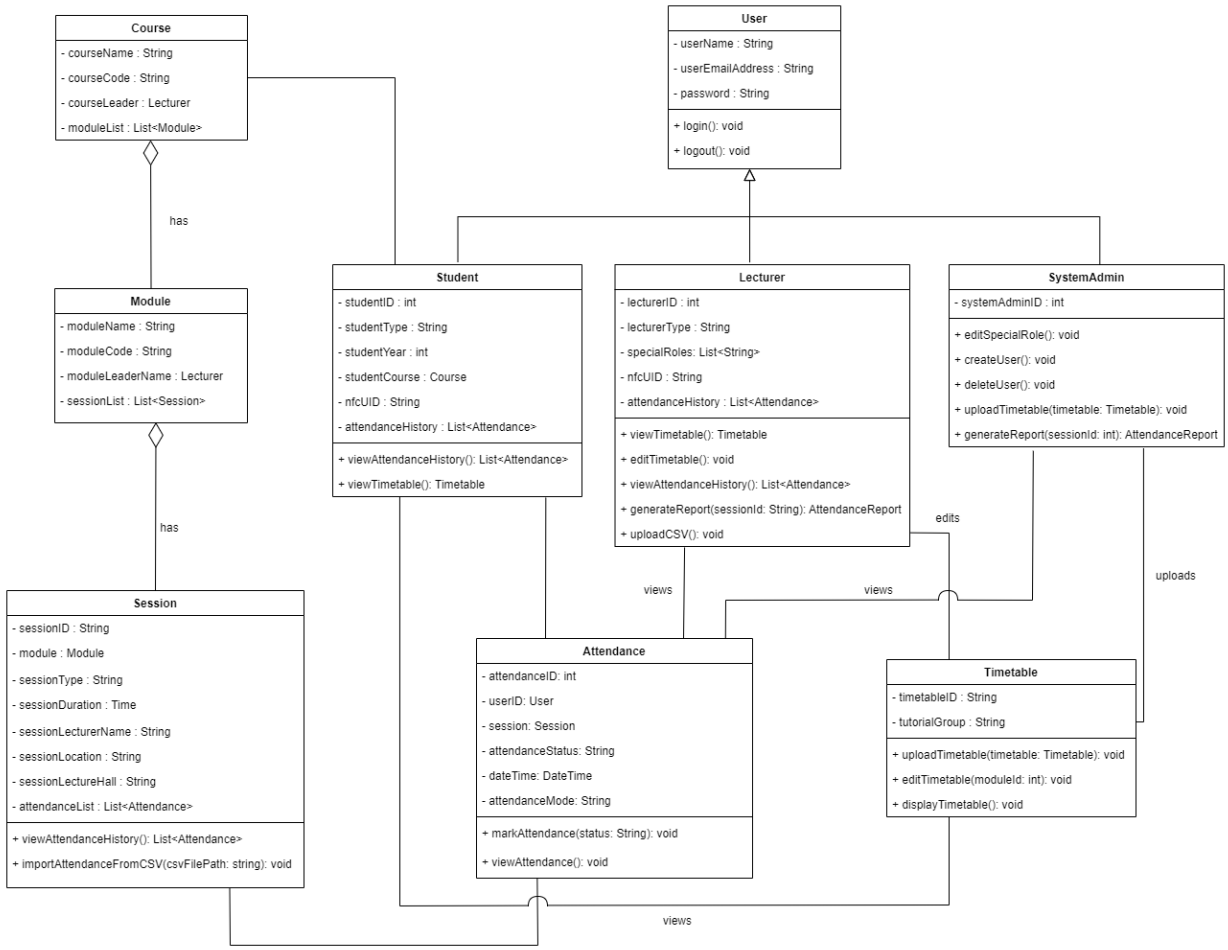


Figure 4 : Class Diagram

6.3.2 Sequence Diagram

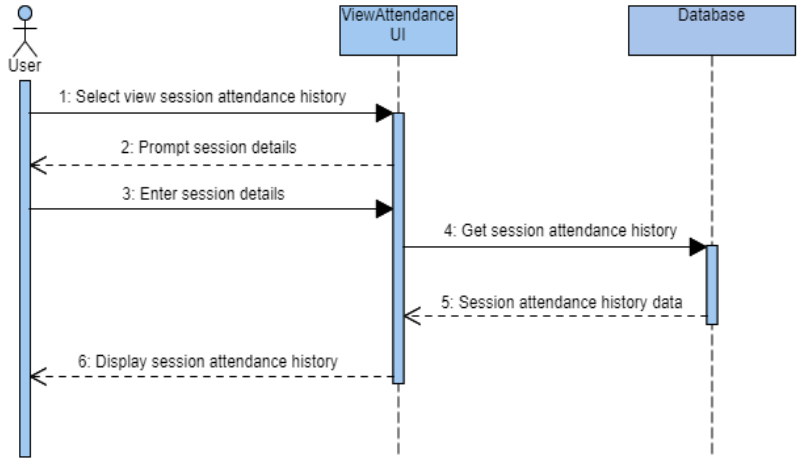


Figure 5 : Sequence Diagram - View Session Attendance History

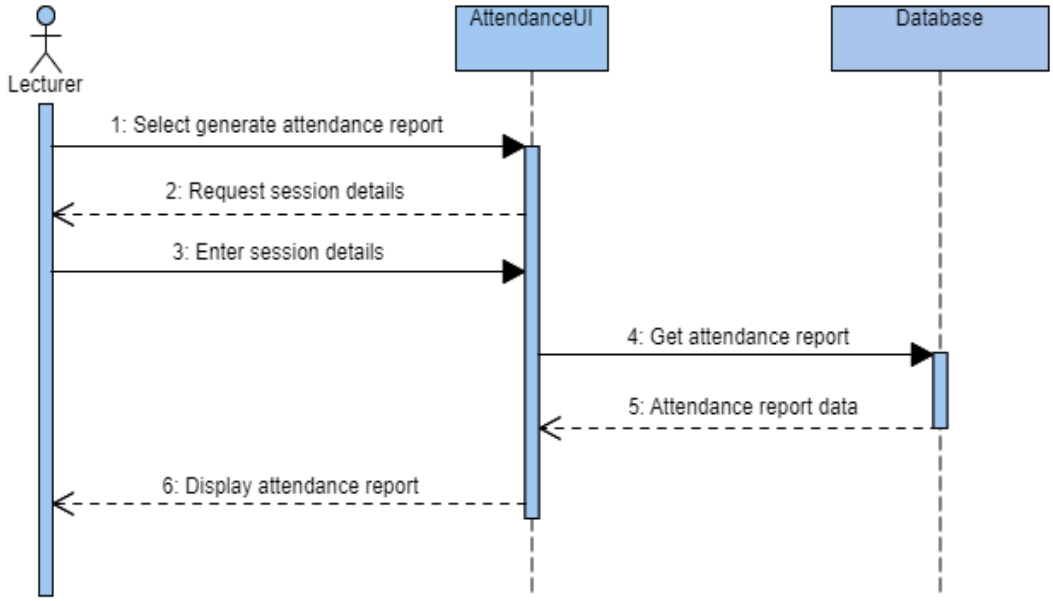


Figure 6 : Sequence Diagram - Generate Report

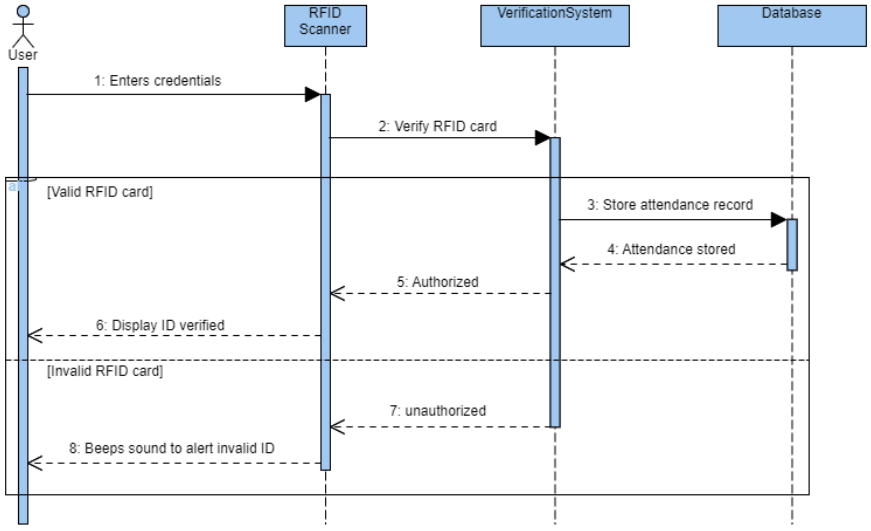


Figure 7 : Sequence Diagram - Mark Attendance

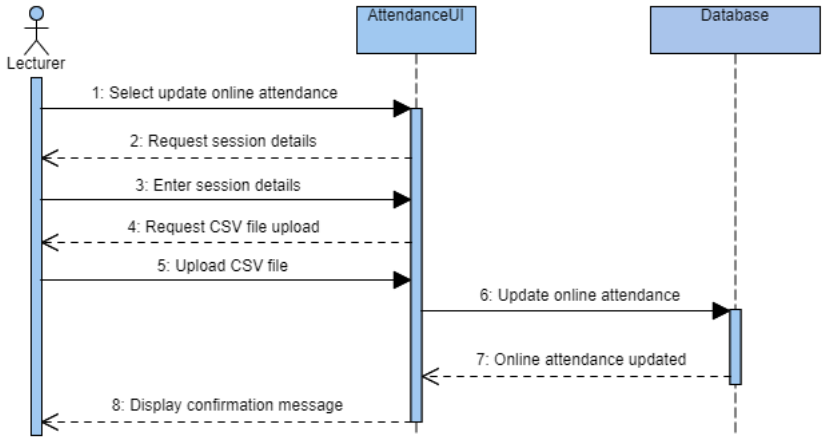


Figure 8 : Sequence Diagram - Update Online Attendance

6.3.3 UI Design

The UI designs for the system were created using Figma. The figure below shows the dashboard and for the lecturer and view attendance history page. The rest of the UI designs can be found under the Appendix Section B.2.

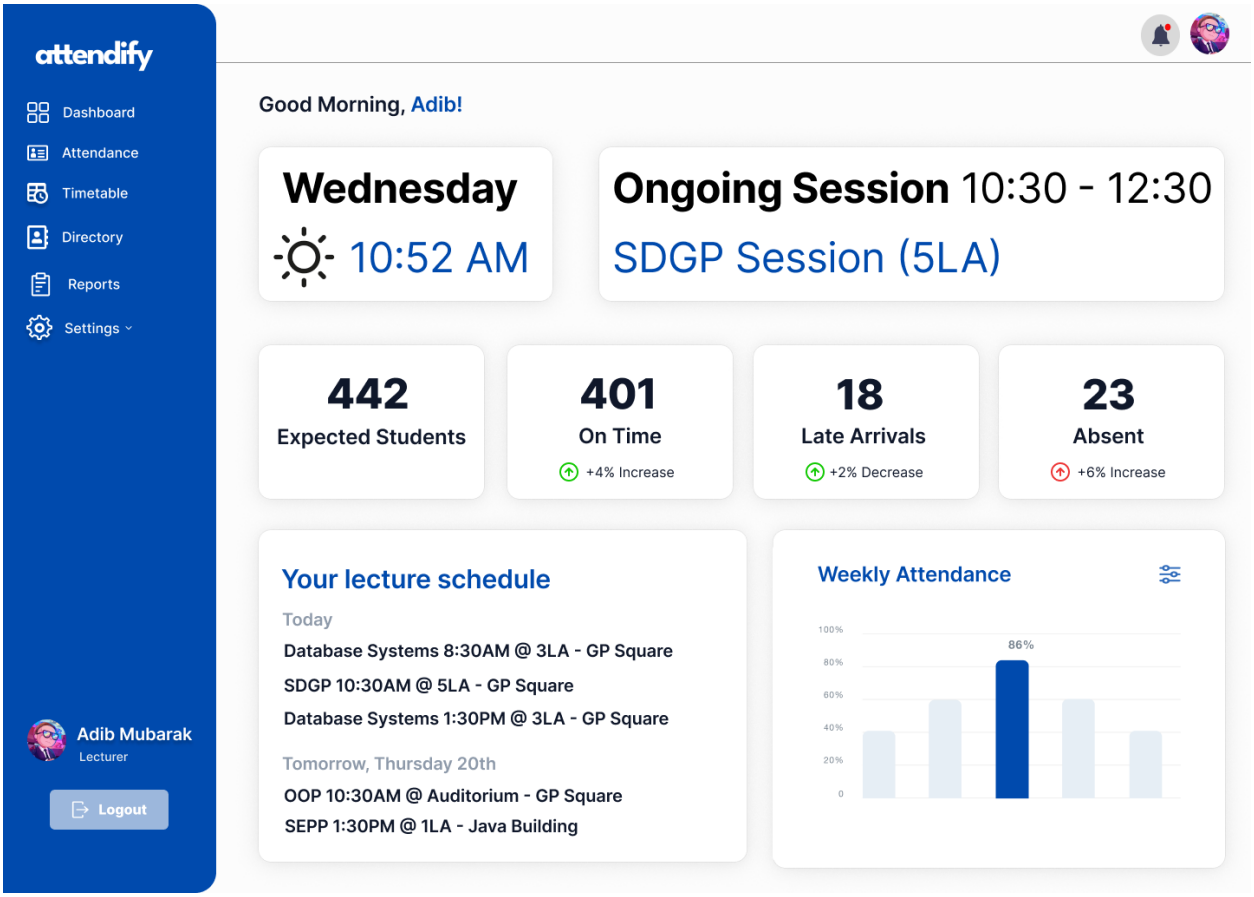


Figure 9 : UI Lecturer Dashboard

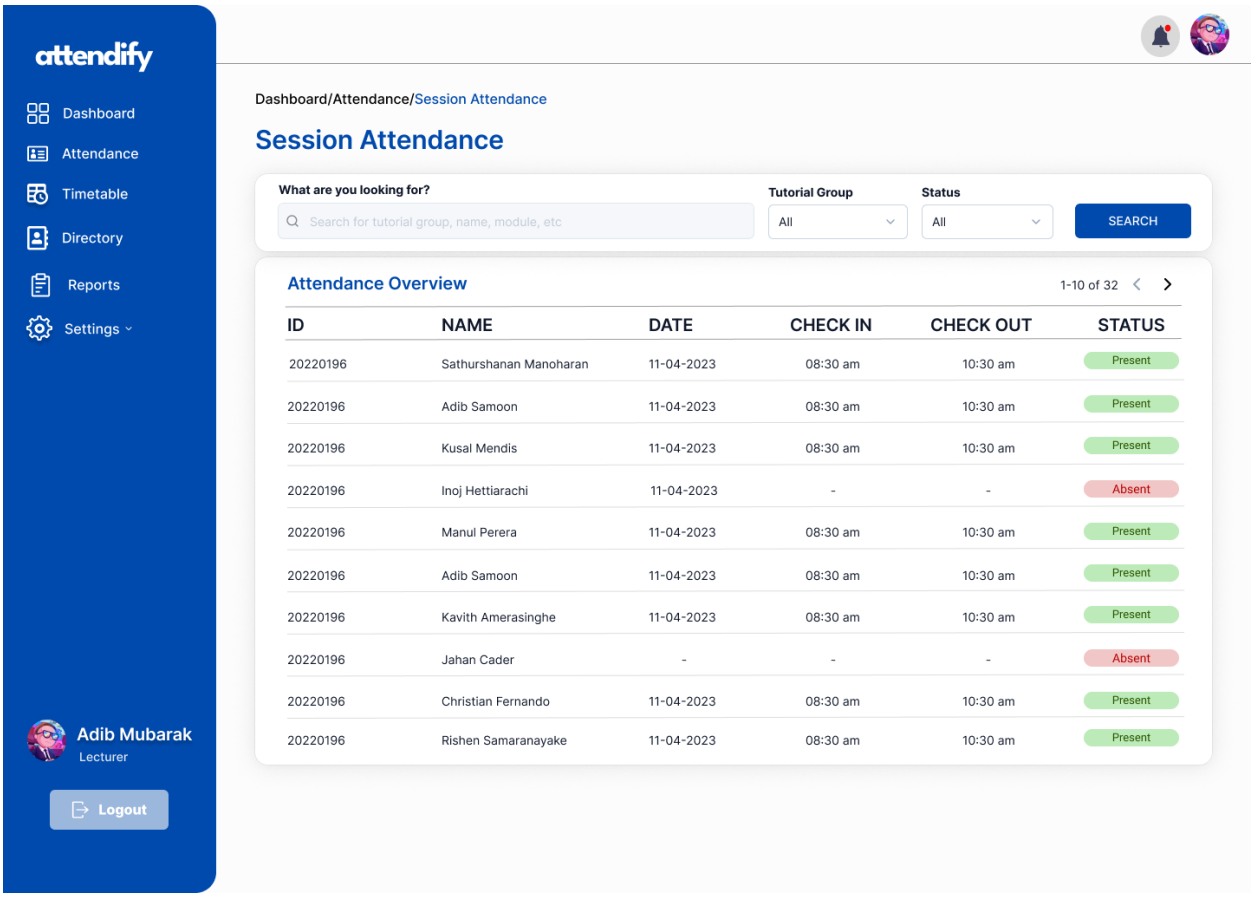


Figure 10 : UI View Session Attendance Page

6.3.4 Activity Diagram

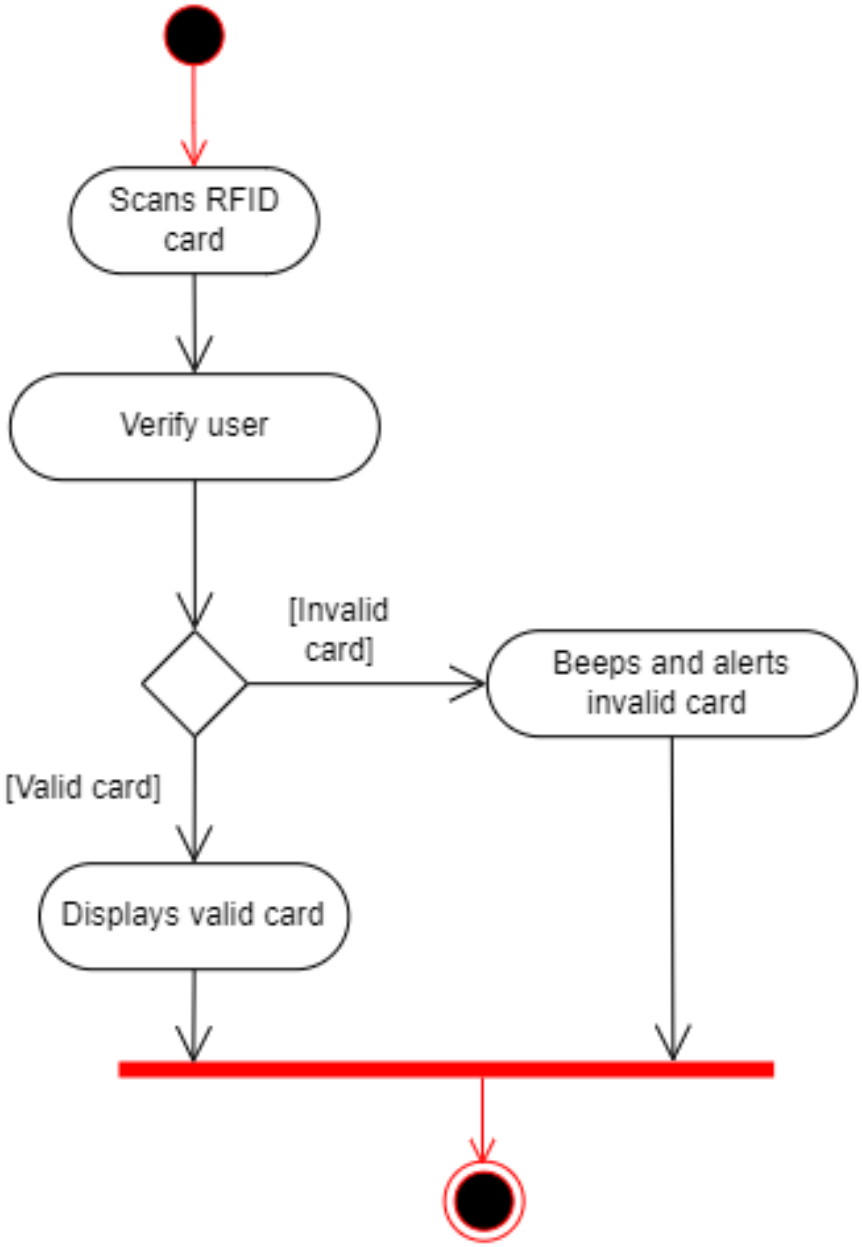


Figure 11 : Activity Diagram - Marking Attendance

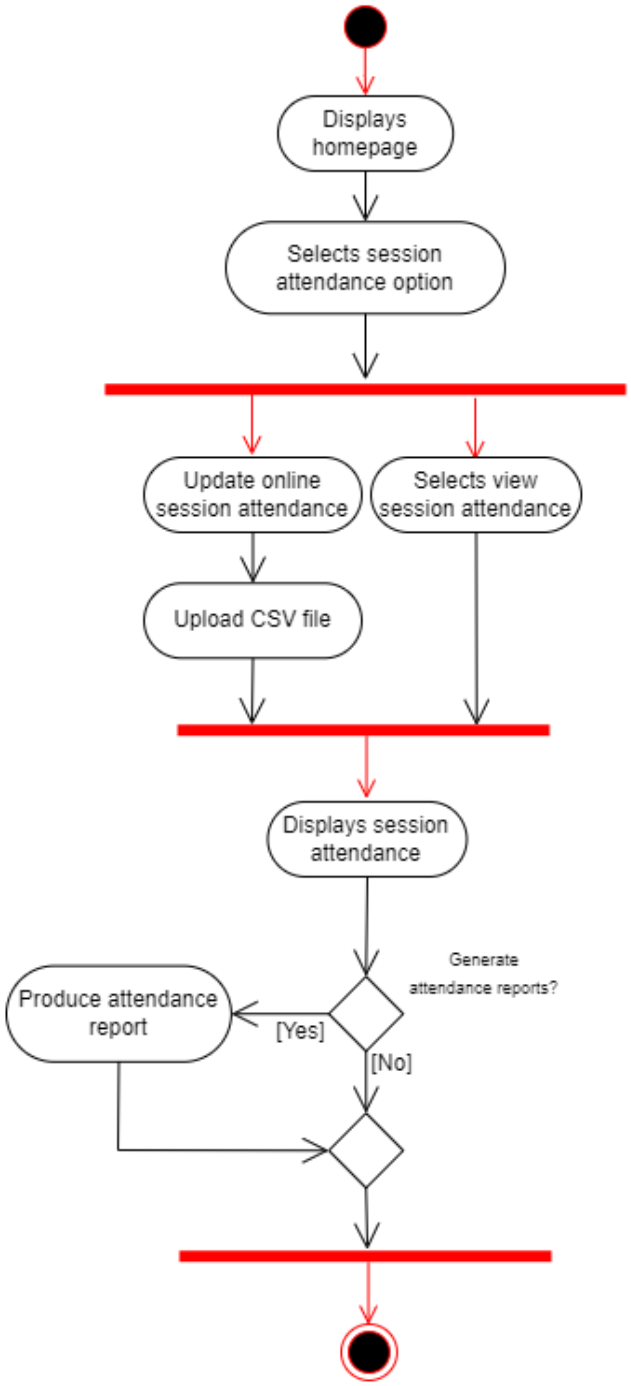


Figure 12 : Activity Diagram - View & Update Attendance

6.4 Chapter Summary

This chapter describes and provides an overview of both the high-level and low-level design of the system. It includes various design elements, such as the high-level architecture diagram, a class diagram, sequence diagrams, UI designs, and activity diagrams. These diagrams serve as aids during the development period. The intention is to utilize these developed diagrams during the implementation phase to ensure the development of a reliable system.

References

BCS Code of conduct (no date). BCS. Available from https://www.bcs.org/membership-and-registrations/become-a-member/bcs-code-of-conduct/ [Accessed 1 January 2024].

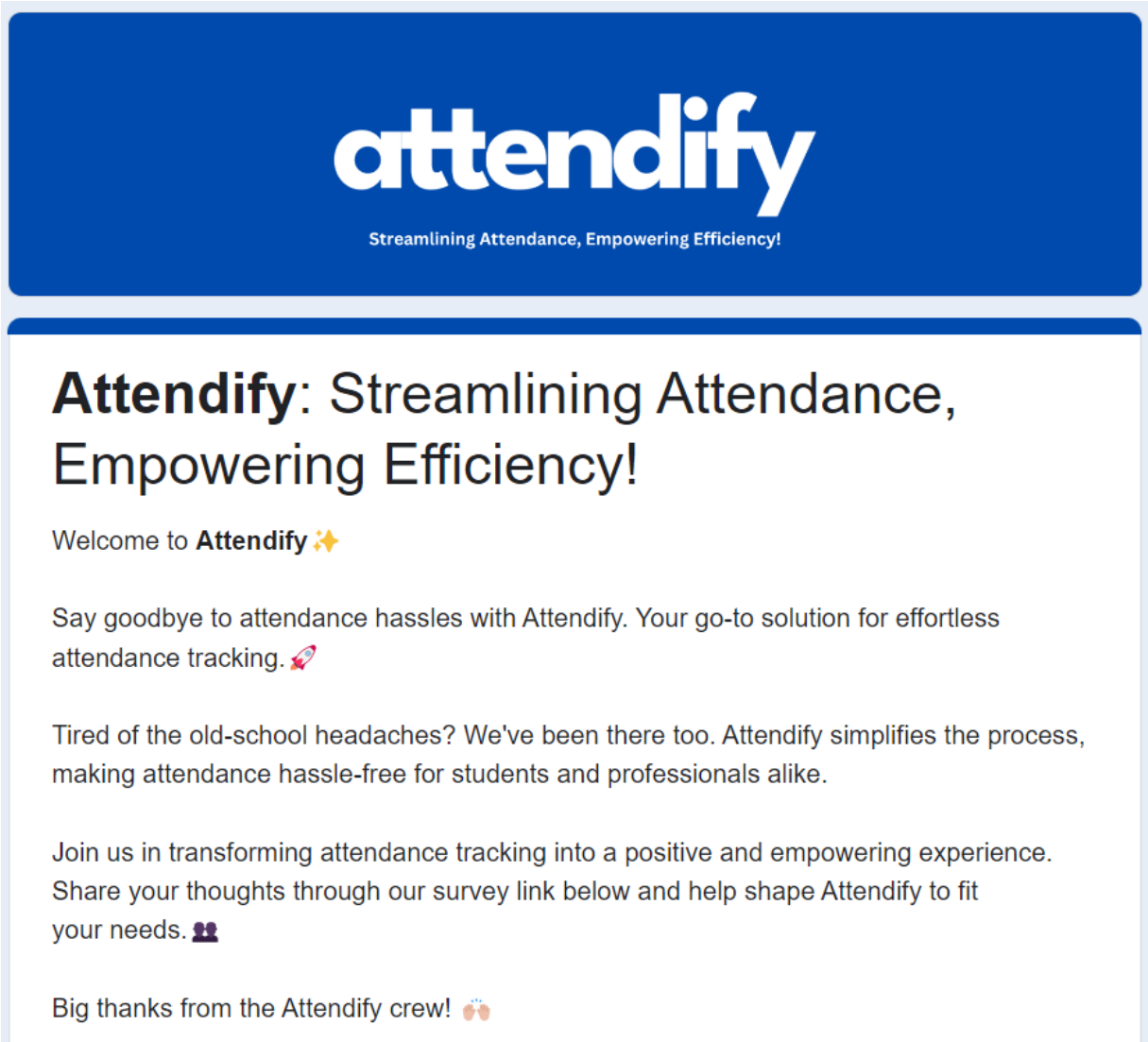
Lecturer. (2024). Interviewee. [Interview]. Online video. Available at: https://drive.google.com/file/d/17btnuU8bAeIZwJvGT9UolKLR04ZygSwt/view?usp=drive_link

MozDevNet (2015). Progressive web apps: MDN. MDN Web Docs. Mozilla. Available from https://developer.mozilla.org/en-US/docs/Web/Progressive_web_apps [Accessed 2 January 2024].

Appendix

Appendix - Section A - System Requirements Specification (SRS)

Appendix Section A.1 - Survey form



Are you satisfied with the current manual (pen and paper method - signature) attendance tracking method? *

- ◯ Yes
- ◯ No

How often do you face difficulties in marking attendance? *

- ◯ Always
- ◯ Often
- ◯ Sometimes
- ◯ Rarely
- ◯ Never

What challenges have you faced with the current manual attendance tracking method, if any?

Your answer

Do you think that automating the attendance tracking process will save time? *

- ◯ Yes
- ◯ No
- ◯ Maybe

What attendance marking method do you prefer? *

- ◯ Pen and Paper method (Signature)
- ◯ NFC Cards
- ◯ Fingerprint
- ◯ Face Recognition
- ◯ Other: _____

Would you prefer a system that allows you to check your attendance record at any time? *

- ◯ Yes
- ◯ No
- ◯ Sometimes

On a scale of 1-10 how important is it to manage attendance to improve your university experience? *

	1	2	3	4	5	6	7	8	9	10	
Not important	○	○	○	○	○	○	○	○	○	○	Extremely important

Would you like to receive automated low attendance warning emails? *

- ○ Yes
- ○ No
- ○ Not sure

Do you face any difficulties in checking your timetable? *

- ○ Always
- ○ Often
- ○ Sometimes
- ○ Rarely
- ○ Never

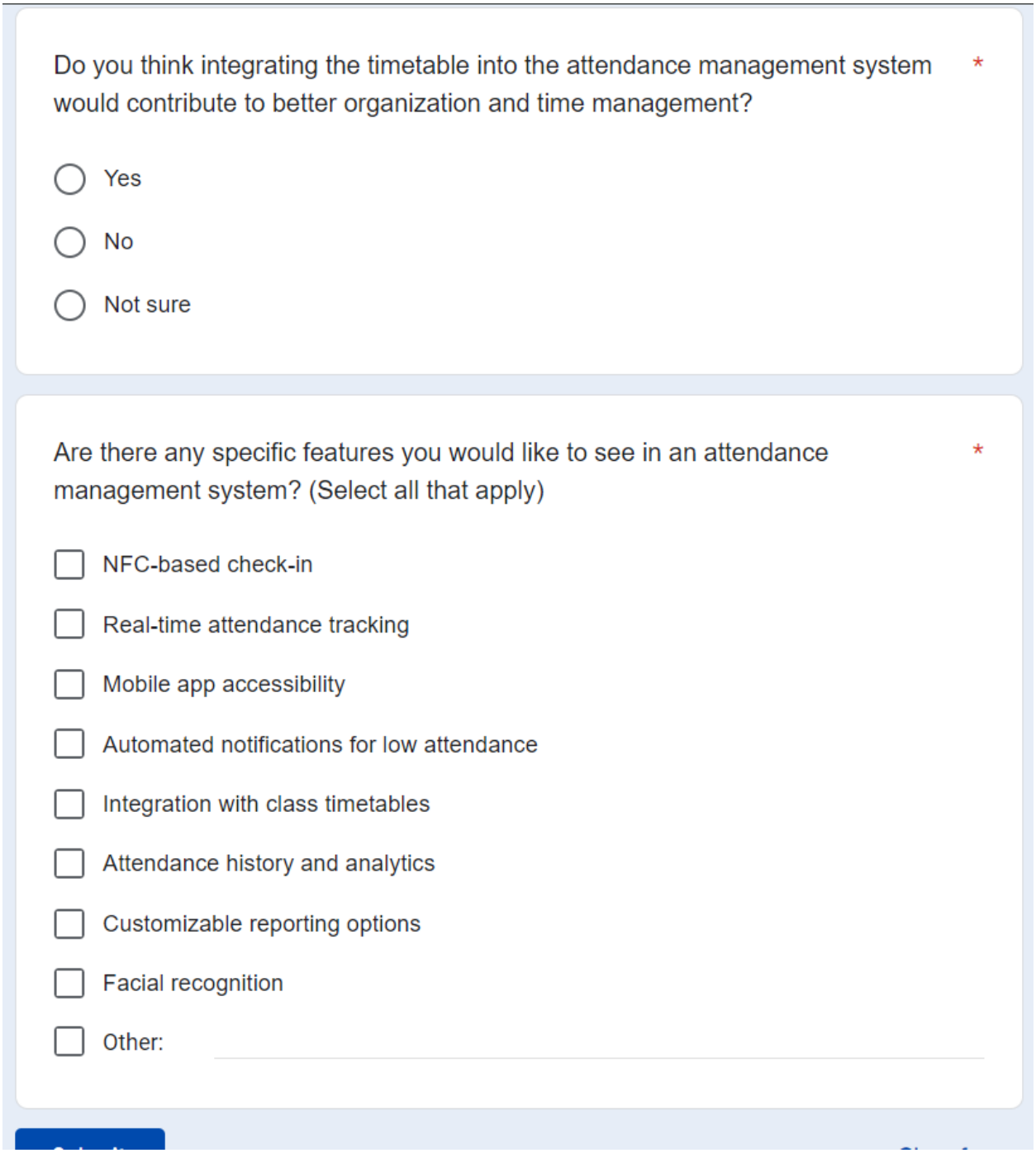
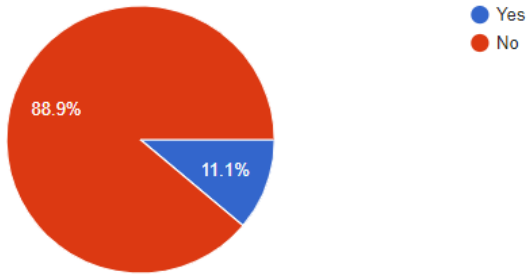
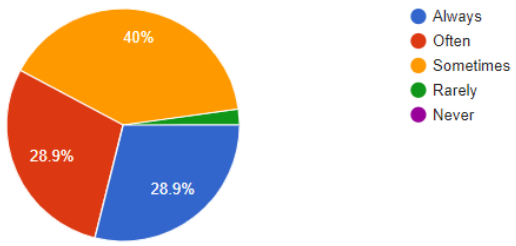


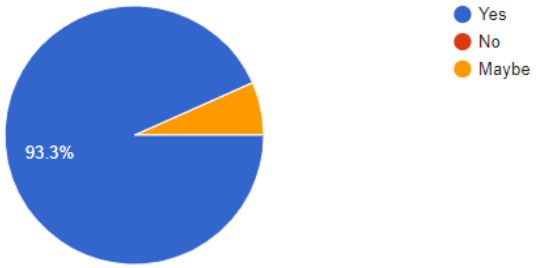
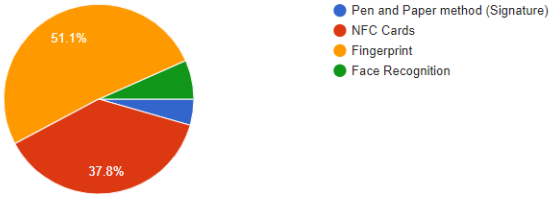
Figure 13 : Survey Questions

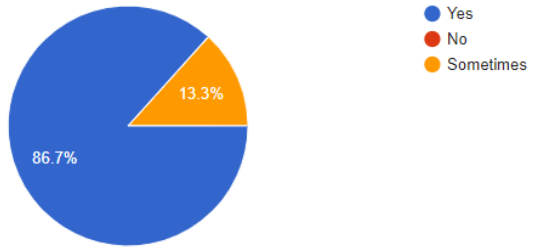
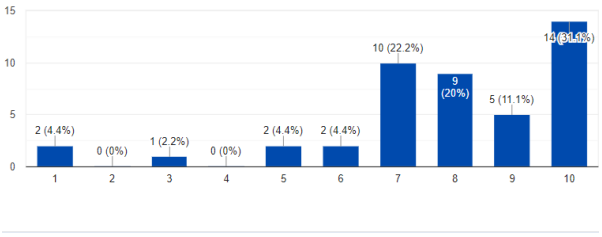
Appendix Section A.2 - Questionnaire analysis

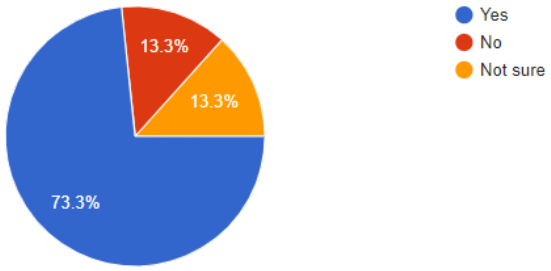
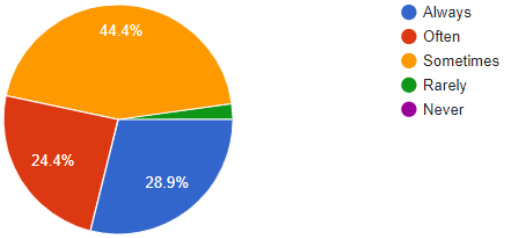
Question 1	Are you satisfied with the current manual (pen

	and paper method - signature) attendance tracking method?
Aim of the Question	Get to know about the students' opinion regarding the current manual attendance tracking method.
	Over 88% of the students are not satisfied with the current manual attendance tracking method. Only 11.1% students are satisfied with the current attendance tracking method.
Conclusion	From this particular question it is evident that students are facing a very bad time when marking attendance manually and therefore a system like Attendify is necessary.
Question 2	How often do you face difficulties in marking attendance?
Aim of the Question	To gain an idea of how frequently students encounter challenges due to the manual attendance tracking method
	More than 28% of the students face difficulty every time when marking attendance. Another 28% of students often face difficulties, and 40% of students sometimes face difficulties.

Conclusion	It is evident that almost everyone has faced difficulties when marking attendance. This signifies the importance of the system that is being proposed.
Question 3	What challenges have you faced with the current manual attendance tracking method, if any?
Aim of the Question	To gain an idea about the challenges that students face due to the manual attendance tracking method.
Common answers: not accurate, Being unable to get the sheet sometimes, Taking a lot of time to find my name on the attendance sheet, Loosing attention (getting disturbed) during a lecture.	Students have been facing issues that could even disrupt the learning process during a lecture.
Conclusion	It is necessary to automate the process of marking attendance to eliminate all these issues.
Question 4	Do you think that automating the attendance tracking process will save time?
Aim of the Question	To get to know about the perspective that students have in mind when it comes to automating the process.

	More than 93% of the students believe that automating the attendance tracking process will save time.
Conclusion	This particular response highlights the need for and significance of Attendify and its automated attendance tracking process.
Question 5	What attendance marking method do you prefer?
Aim of the Question	To get to know the preferred method that the students want to use to mark attendance.
	51% of the students want a finger print-based solution, and 37% of the students want a card-based attendance system.
Conclusion	Due to several reasons, such as high cost to implement and maintain, time-consuming implementation, etc., the team has decided that going with an RFID/NFC-based attendance system is better.
Question 6	Would you prefer a system that allows you to check your attendance record at any time?
Aim of the Question	To gain an idea of whether the students want

	to check their attendance history.
	86% of the students want a feature where they can view their attendance history.
Conclusion	In order to achieve this particular feature, the attendance tracking must be automated and saved in a database.
Question 7	On a scale of 1-10 how important is it to manage attendance to improve your university experience?
Aim of the Question	This question was given to check the student's mentality regarding attendance, which could play a major role in attendidy's success.
	More than 84% of the students believe that managing attendance is an important part of their day to day lives.
Conclusion	It is important to have an attendance history for every student that they can access at any time.
Question 8	Would you like to receive automated low attendance warning emails?

Aim of the Question	To get an understanding of whether the students want to be updated regarding low attendance.
	73% of students want automated low-attendance warning emails. Of the remaining 26%, 13% are not sure, and the other 13% do not want this feature.
Conclusion	A conclusion was made that students want to receive an update regarding their lower attendance. This particular feature was decided to be added to the future scope.
Question 9	Do you face any difficulties in checking your timetable?
Aim of the Question	
	More than 28% of students always face trouble checking their timetable. 24% of the students often face challenges when checking their timetables. 44% of the students occasionally face trouble when checking their timetable.
Conclusion	The conclusion was that an integrated timetable is necessary.
Question 10	Do you think integrating the timetable into the attendance management system would

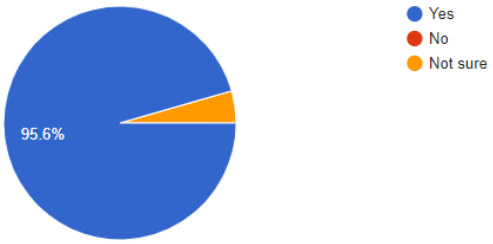
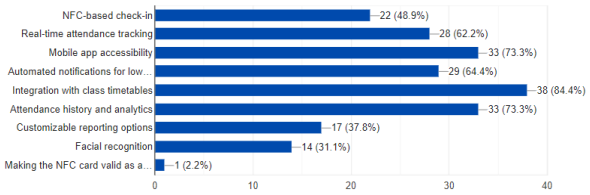
	contribute to better organization and time management?
Aim of the Question	To know whether the students want the timetable to be integrated into the system.
	More than 95% of the students want to have an integrated timetable with the attendance management system.
Conclusion	It is important to have this feature as a high priority because almost everyone has faced challenges when checking their timetables.
Question 11	Are there any specific features you would like to see in an attendance management system?
Aim of the Question	To get an idea of the features that the students are expecting in this system.
	More than 84% of students want the timetable integrated with the attendance system. More than 62% of students want a real time attendance tracking feature.
Conclusion	All these suggestions given by the students were taken into consideration when making the functional requirements.

Table 9 : Questionnaire Analysis

Appendix Section A.3 - Interview analysis

Question 1	What are the main challenges or pain points with the current attendance process?
Aim of the Question	To get an idea of the problems that lecturers face due to the current attendance method.
Observation	• The main problem is that the marking of attendance is paper based. • The attendance sheet does not come on time. • There was not enough time for the students to sign in for the large class. • Lecturers have to wait until every student signs the attendance sheet.
Conclusion	In conclusion, the current attendance process has created several challenges for lecturers, such as delays in receiving the paper, not enough time for the students to mark attendance, and lecturers need to wait after a particular lecture finishes. A solution that could resolve all these problems would be ideal.
Question 2	Are there specific features or functionalities that would enhance the usability of the system for you?
Aim of the Question	To gain an understanding regarding the features that are needed from a lecturer's perspective.
Observation	• Being able to monitor who comes into a lecture would be a helpful feature.
Conclusion	A feature is needed to track and identify the

	attendees in real time.
Question 3	What do you think about the accuracy of manual attendance records, and have there been instances where discrepancies needed resolution?
Aim of the Question	To get an idea about the accuracy of the attendance.
Observation	• Students might forget to mark attendance if they do not get the paper. • Sometimes more signatures are on the sheet than the number of students in the session. • Students are marking proxy attendance.
Conclusion	Marking the attendance before entering the class would save the lecturer's and student's time immensely.
Question 4	Do you have specific preferences regarding notifications or alerts related to student attendance, and how would you prefer to be informed of irregularities?
Aim of the Question	To determine if lecturers would like to have any additional features in the system that are related to tracking attendance.
Observation	• Anything that can be sent through push notifications will be helpful.
Conclusion	Since Attendify is a web-based solution, the team decided to build the solution using Progressive Web Apps (PWA) technology (MozDevNet, 2015).

Question 5	How important is it for the system to provide feedback or insights on attendance trends, and are there specific analytics or reports that would be particularly useful?
Aim of the Question	To determine if lecturers would like to get any reports regarding attendance for any particular session.
Observation	• Analytical reports would benefit the lecturers a lot because they have a visual representation of the information.
Conclusion	The Attendify team has already decided to give access to the lectures to generate reports based on sessions or modules.
Question 6	How do you think an RFID-based attendance system would impact student attendance?
Aim of the Question	To understand whether lecturers are satisfied to go ahead with an RFID based attendance tracking system.
Observation	• Having an RFID based solution will almost eliminate all the issues mentioned, but the development team needs to figure out a way to reduce proxy marking.
Conclusion	In conclusion, the decision was made to come up with a protection layer to eliminate the issue of proxy marking that may arise.

Question 7	What features would you like to see if a timetable module is implemented within the system?
Aim of the Question	To determine whether this feature would benefit the lecturers.
Observation	• Push notifications when there is a lecture is helpful. • If timetables can be integrated with google calendar, that's useful.
Conclusion	The team decided to integrate the system with google calendar as a future work.

Table 10 : Interview Analysis

Appendix Section A.4 - Use case descriptions

Use Case Name	Login
Use Case ID	UC-001
Description	User logging into the system
Priority	High
Primary Actor	Lecturer, Student, Admin
Supporting Actors	None
Pre-Conditions	User should have a proper internet connection and should be registered under the particular organisation.

Trigger	The user wants to login to the system.	
Main flow	**Actors**	**System**
	1) User opens the web application 2) User enters the credentials	3) System validates the credentials entered 4) System redirects the user to the desired landing page
Exception flow	**Actors**	**System**
	None	None
Alternate flow	**Actors**	**System**
	1) User enters invalid username or password 3) User enters valid username or password	2) System displays incorrect username or credentials 4) System redirects the user to the desired landing page
Exclusions	None	
Post Conditions	The system displays the desired landing page.	

Table 11 : Use Case description – Login

Use Case Name	View personal attendance
Use Case ID	UC-003
Description	Users viewing their personal attendance history

Priority	High	
Primary Actor	Student	
Supporting Actors	None	
Pre-Conditions	Student should be logged in to the system.	
Trigger	The student wants to view their personal timetable	
Main flow	**Actors**	**System**
	1) Student selects view attendance history option	2) System displays their personal attendance history
Exception flow	**Actors**	**System**
	1) Student selects view attendance history option 3) Student connects to internet	2) System does not work due to bad internet connection 4) System displays their personal attendance history
Alternate flow	**Actors**	**System**
	1) Student selects view attendance history option	2) System displays no attendance history
Exclusions	None	
Post Conditions	The system displays the desired student's personal attendance history.	

Table 12 : Use Case description - View Personal Attendance

Use Case Name	View personal timetable	
Use Case ID	UC-002	
Description	Users viewing their personal timetables	
Priority	High	
Primary Actor	Lecturer, Student	
Supporting Actors	None	
Pre-Conditions	User should be logged in to the system.	
Trigger	The user wants to view their personal timetable	
Main flow	**Actors**	**System**
	1) User selects view timetable option	2) System displays the desired timetable
Exception flow	**Actors**	**System**
	1) User selects view timetable option 3) User connects to internet	2) System does not work due to bad internet connection 4) System displays the desired timetable
Alternate flow	**Actors**	**System**
	None	None

Exclusions	None
Post Conditions	The system displays the desired timetable.

Table 13 : Use Case description - View Personal Timetable

Use Case Name	Upload timetable	
Use Case ID	UC-005	
Description	Admin uploading the timetable to the system	
Priority	High	
Primary Actor	Admin	
Supporting Actors	None	
Pre-Conditions	Admin should have the correct format to upload the timetable.	
Trigger	The admin wants to upload the timetable.	
Main flow	**Actors**	**System**
	1) Admin selects upload timetable option 4) Admin uploads the timetable	2) System displays the upload timetable option page 3) System prompts the admin to upload the timetable 5) System updates the new timetable
Exception flow	**Actors**	**System**

Alternate flow	**Actors**	**System**
	1) Admin selects upload timetable option 4) Admin uploads the timetable in an invalid format 6) Admin uploads in a valid format	2) System displays the upload timetable option page 3) System prompts the admin to upload the timetable 5) System alerts invalid format uploaded 7) System updates the new timetable
Exclusions	None	
Post Conditions	The system updates the new timetable.	

Table 14 : Use Case description - Upload Timetable

Use Case Name	Update user
Use Case ID	UC-006
Description	Admin updating a user
Priority	High
Primary Actor	Admin
Supporting Actors	Student, Lecturer
Pre-Conditions	Admin should be logged into the system.

Trigger	The admin wants to update a user.	
Main flow	**Actors**	**System**
	1) Admin selects update user option 4) Admin enters the desired details	2) System displays the update user option page 3) System prompts the admin to enter the details 5) System updates the user
Exception flow	**Actors**	**System**
	1) Admin selects update user option 3) Admin connects to the internet	2) System does not work due to bad internet connection 4) System continues to work in the desired manner
Alternate flow	**Actors**	**System**
	None	None
Exclusions	None	
Post Conditions	The system creates, updates or deletes users according to the details admin gave.	

Table 15 : Use Case description - Update User

Use Case Name	Edit timetable
Use Case ID	UC-008

Description	Lecturer has the access to edit the time slot in the timetable which is under their responsibility.	
Priority	High	
Primary Actor	Lecturer	
Supporting Actors	Admin	
Pre-Conditions	Lecturer should be logged into the system.	
Trigger	The lecturer wants to edit the timetable.	
Main flow	**Actors**	**System**
	1) Lecturer selects edit timetable option 4) Lecturer selects a desired option 6) Lecturer updates the necessary details	2) System displays the edit timetable option page 3) System prompts the lecturer to choose a desired option to make an edit 5) System asks the lecturer to update the time slot
Exception flow	**Actors**	**System**
	1) Lecturer selects edit timetable option 3) Lecturer connects to the internet	2) System does not work due to bad internet connection 4) System continues to work in the desired manner
Alternate flow	**Actors**	**System**

	None	None
Exclusions	Schedule additional lectures, Cancel lecture	
Post Conditions	The system updates the timetable.	

Table 16 : Use Case description - Edit Timetable

Appendix - Section B - System Architecture & Design

Appendix Section B.1 - Sequence Diagrams

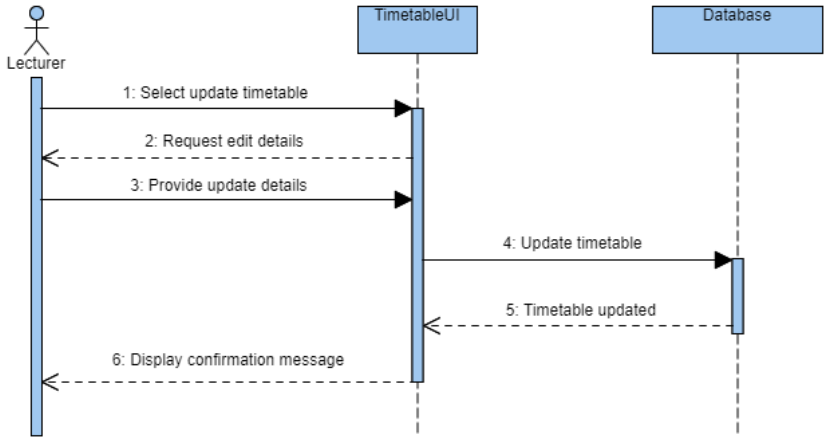


Figure 14 : Sequence Diagram - Edit Timetable

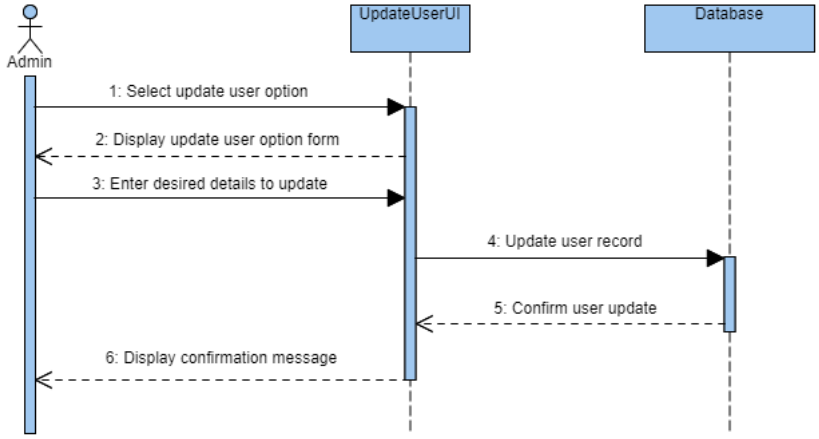


Figure 15 : Sequence Diagram - Update User

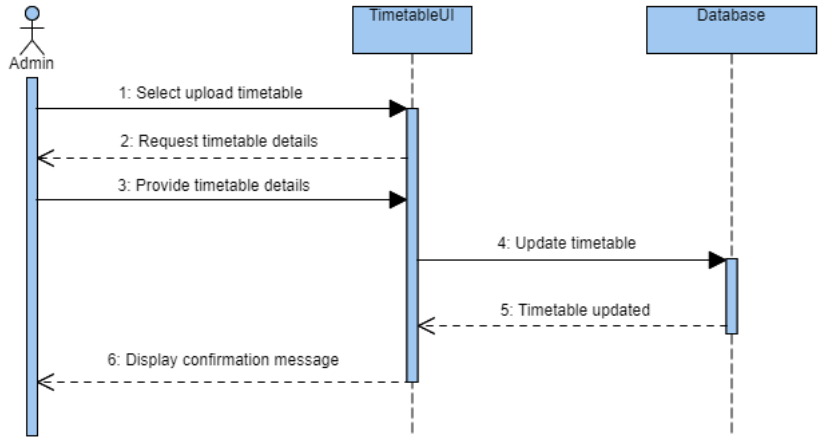


Figure 16 : Sequence Diagram - Upload Timetable

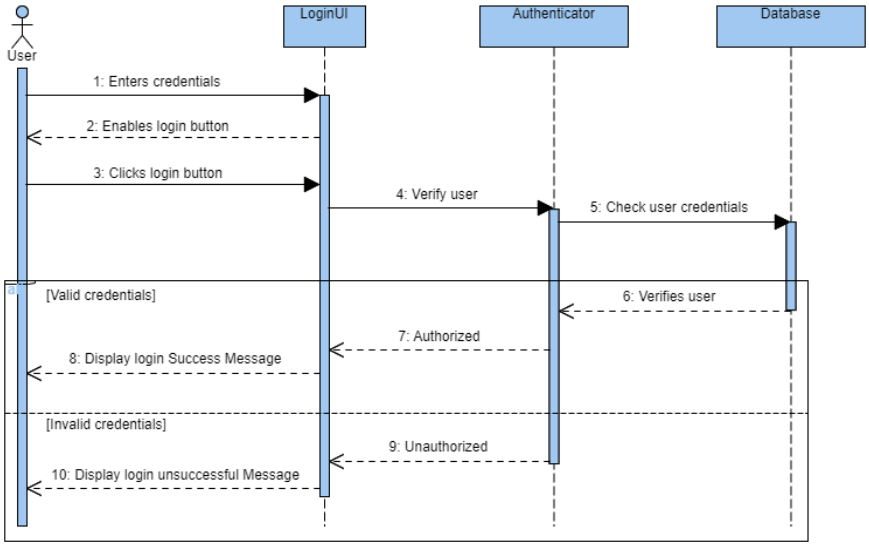


Figure 17 : Sequence Diagram - Login

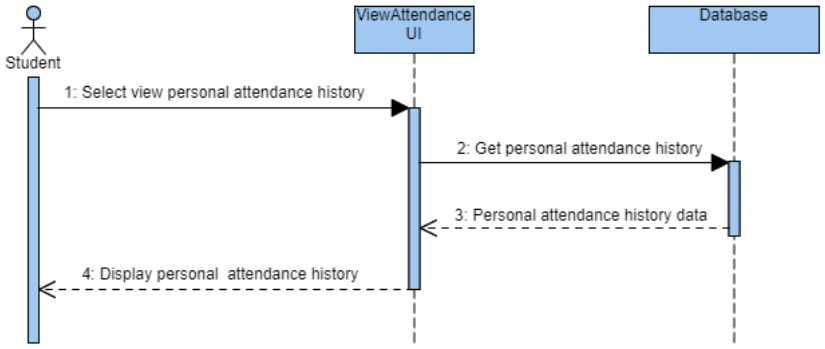


Figure 18 : Sequence Diagram - View Personal Attendance

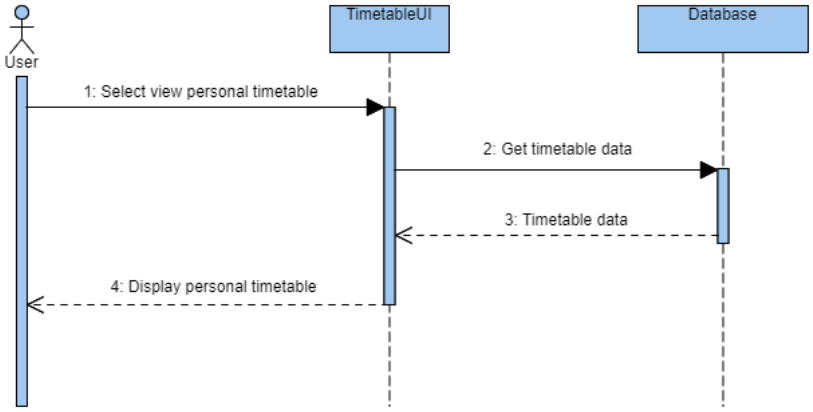


Figure 19 : Sequence Diagram - View Personal Timetable

Appendix Section B.2 - UI Designs

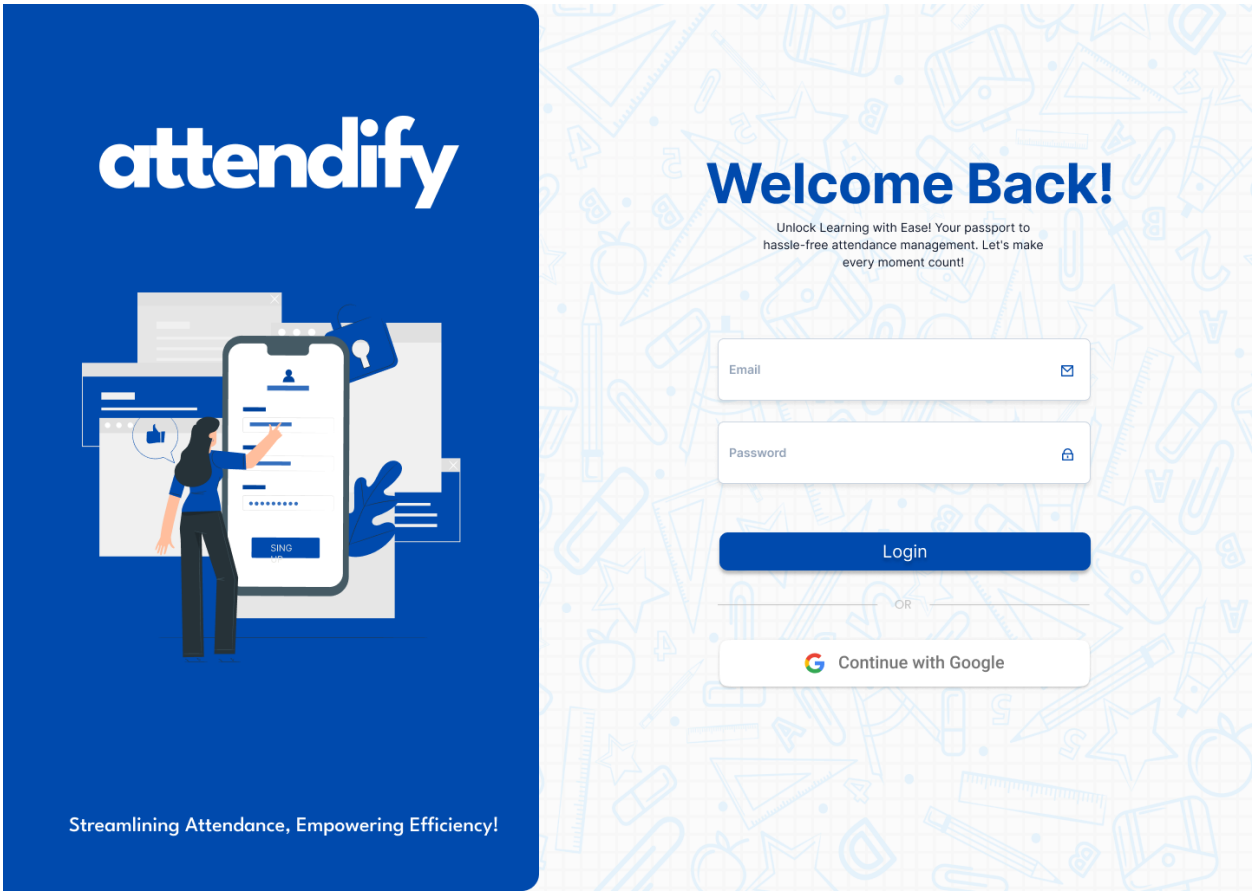


Figure 20 : UI Login Page

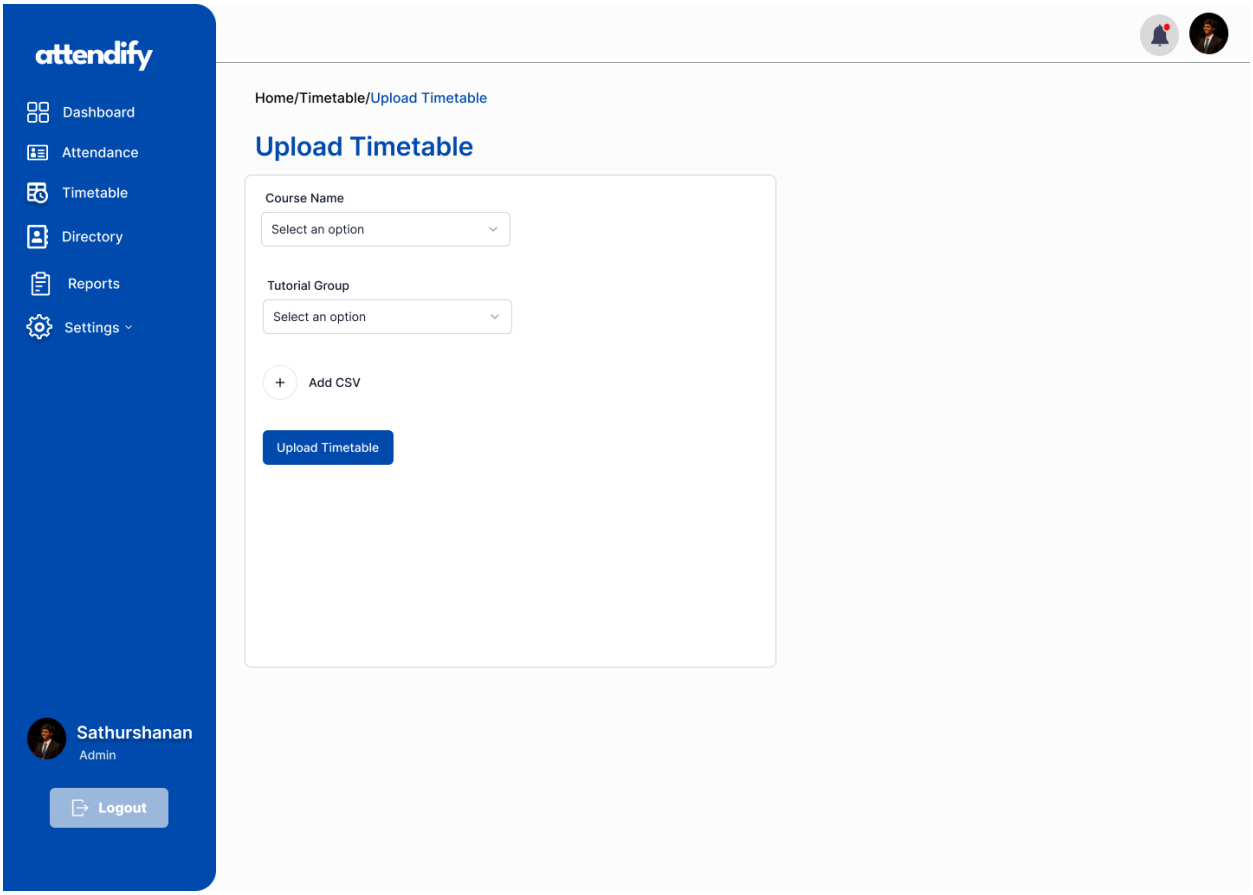


Figure 21 : UI Upload Timetable Page for Admin

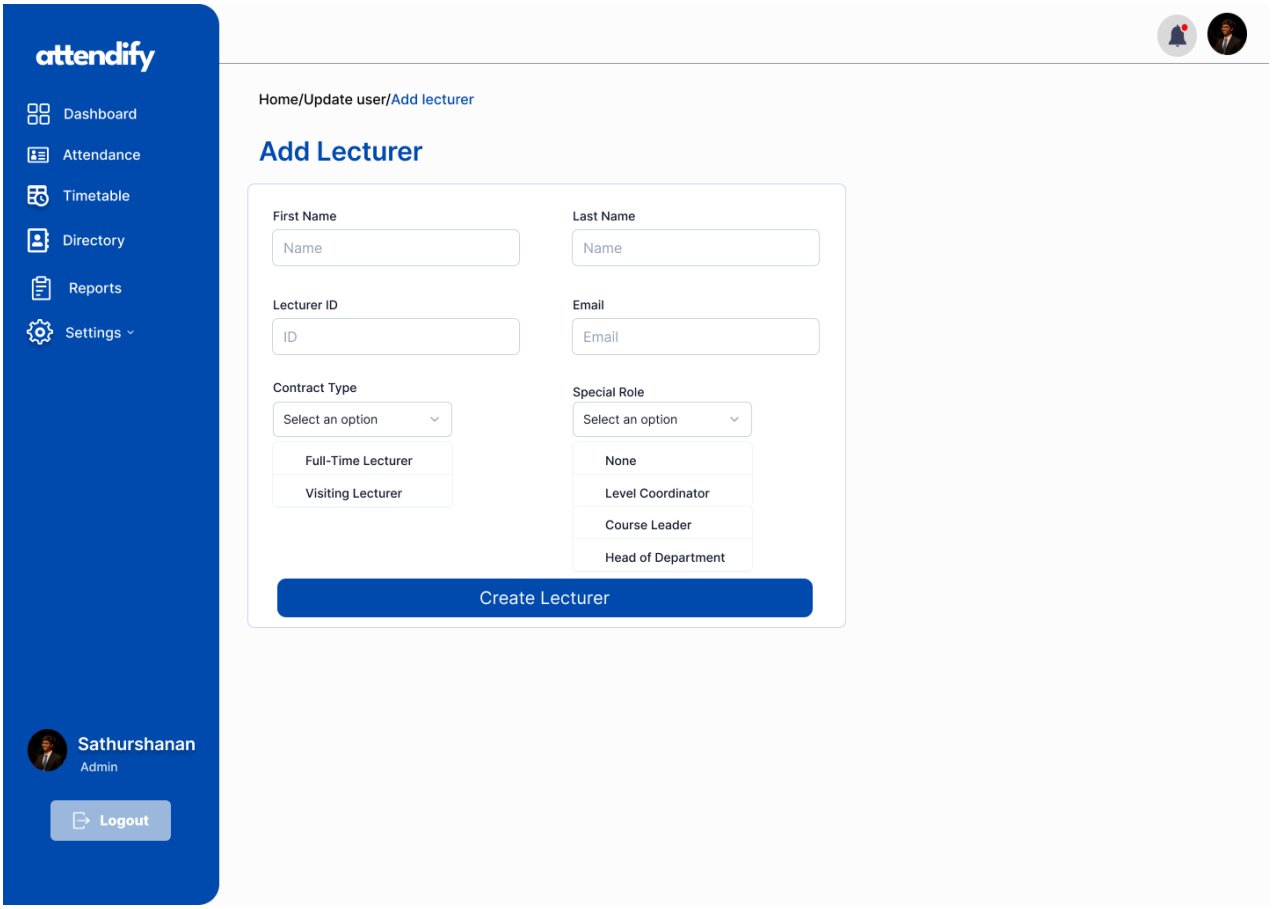


Figure 22 : UI Add Lecturer Page for Admin

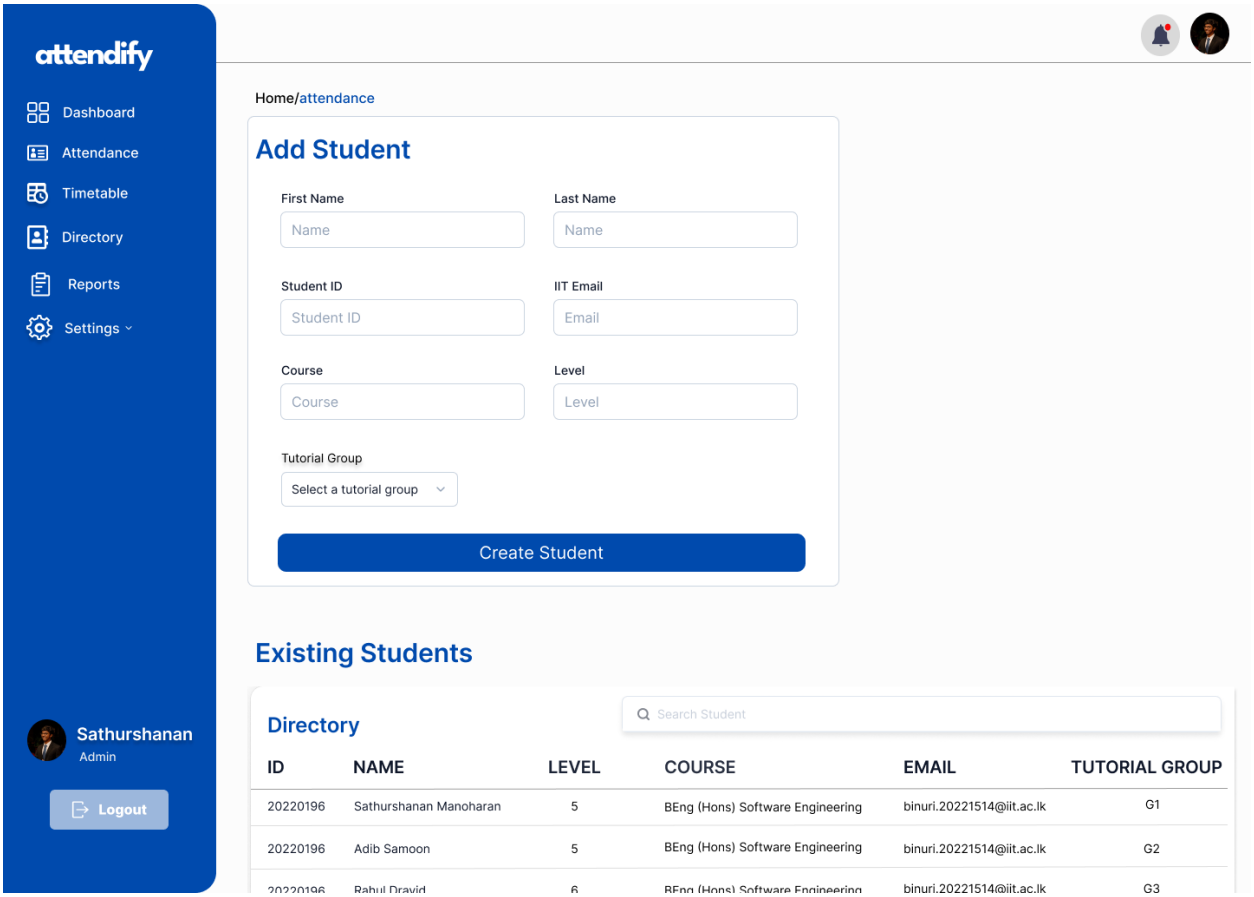


Figure 23 : UI Add Student Page for Admin

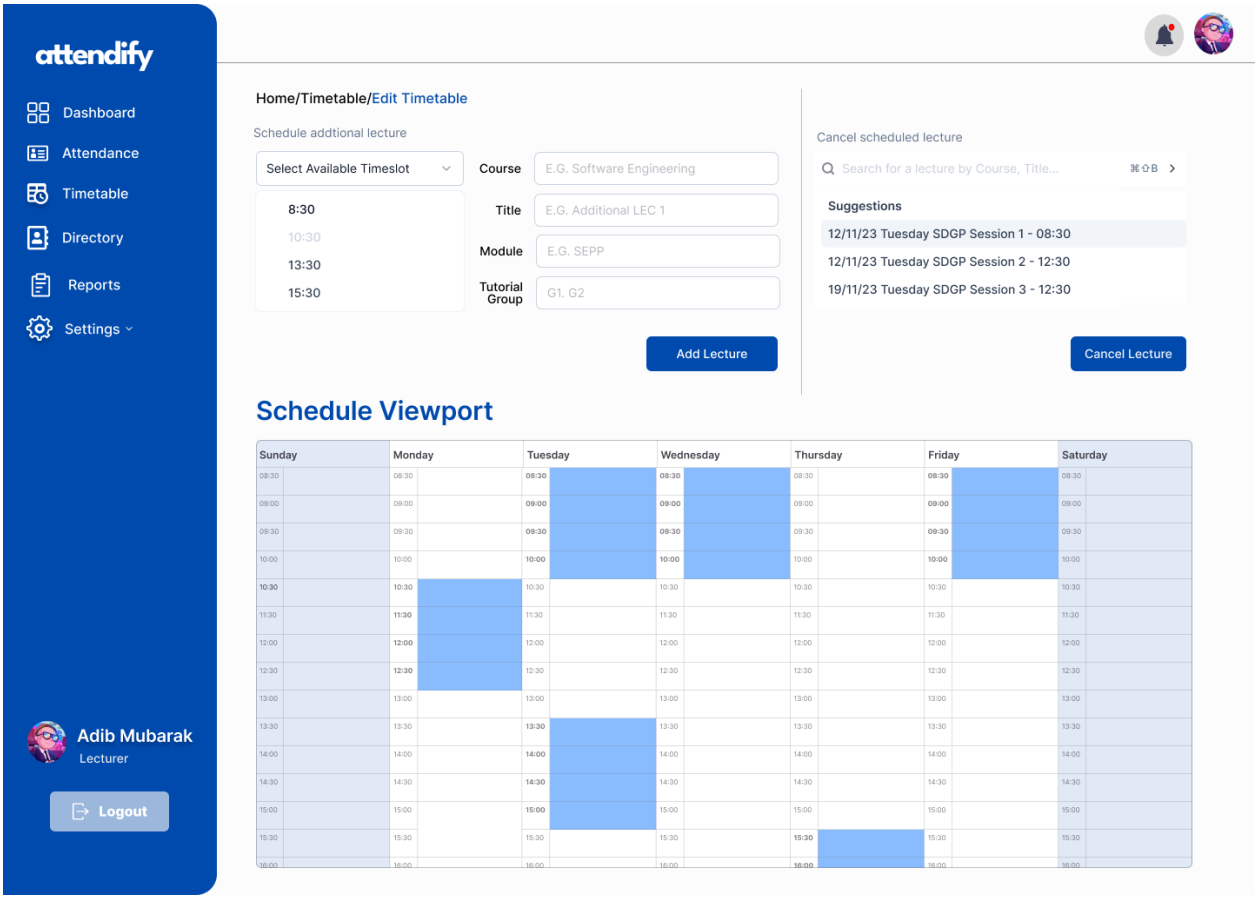


Figure 24 : UI View & Edit Timetable Page for Lecturer

Appendix Section B.3 - Activity Diagrams

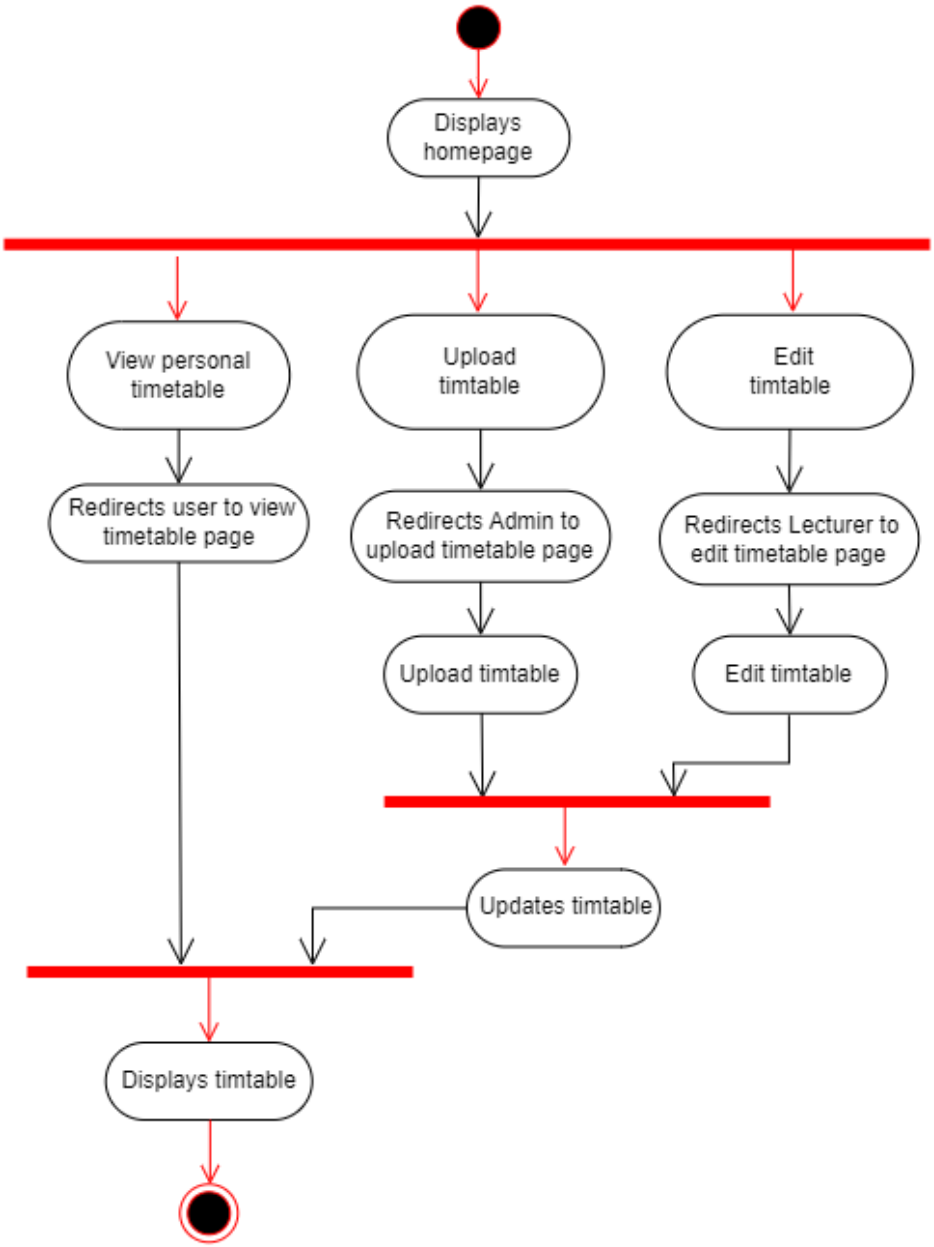


Figure 25 : Activity Diagram - Timetable flow

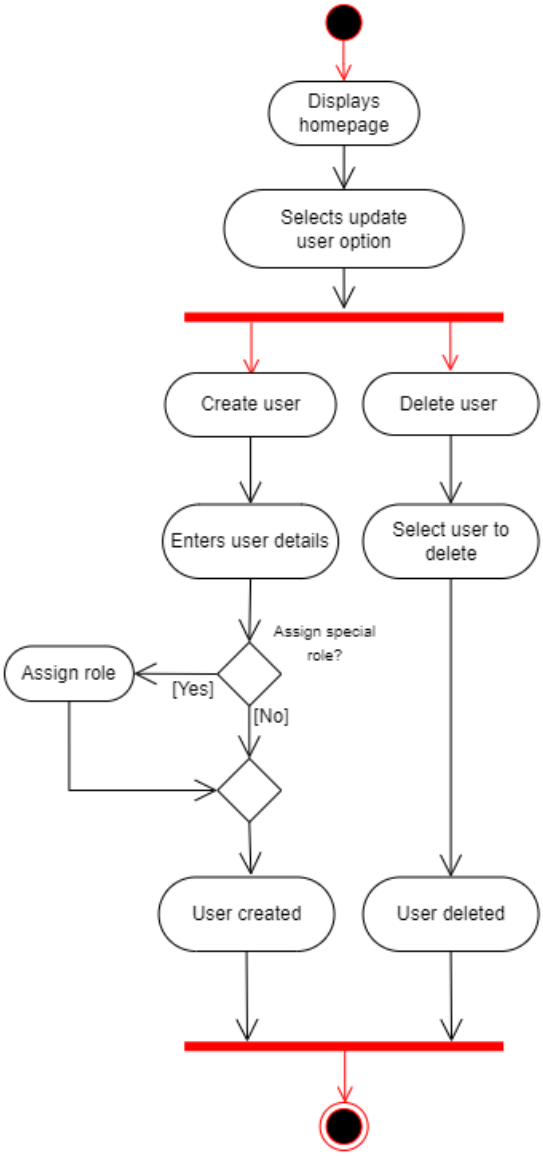


Figure 26 : Activity Diagram - Update User

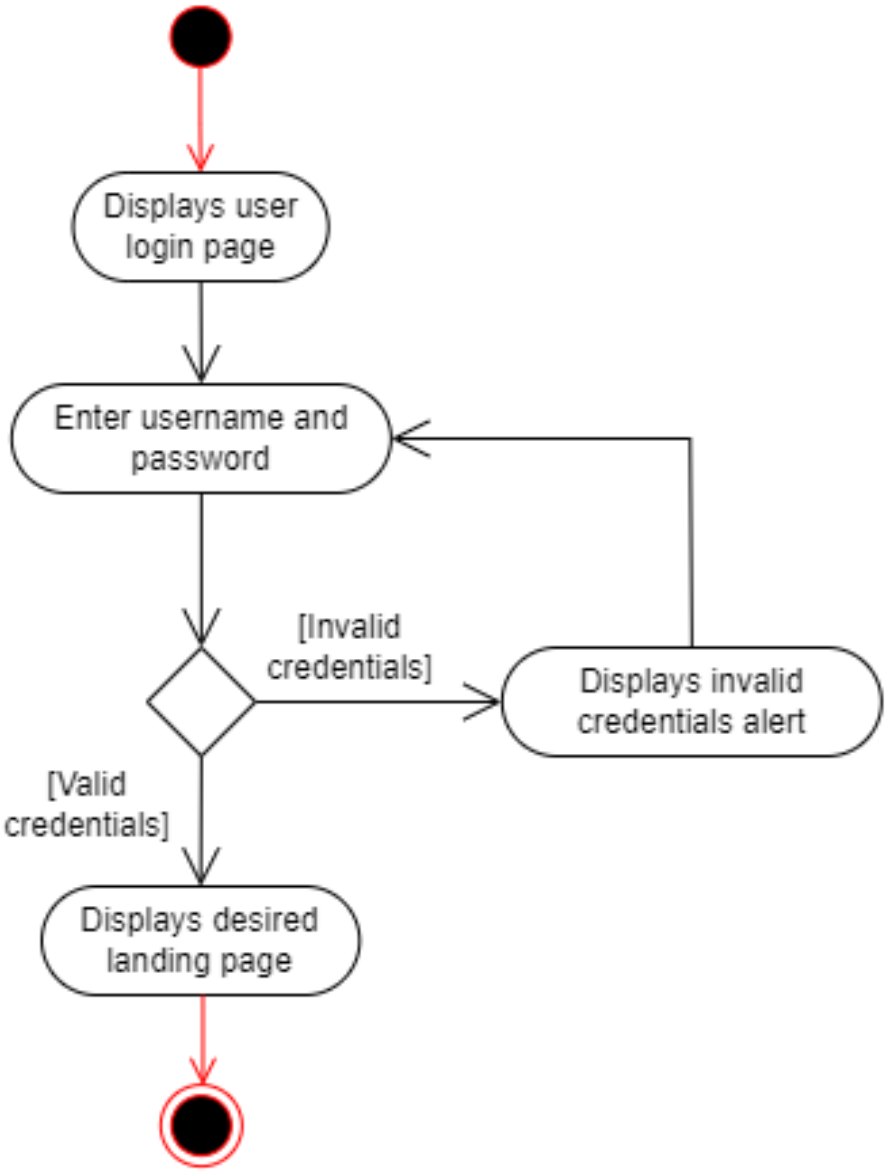


Figure 27 : Activity Diagram - Login